Deployment

English version of ../../operations/deployment.md.

Options

Option	File	Note
Docker Compose (recommended)	`docker-compose.yml`	one service, the `brainstorm-data` volume
Container by hand	`Dockerfile`	multi-stage, non-root `appuser`
Process manager	`Procfile`, `gunicorn.conf.py`	`gunicorn -c gunicorn.conf.py wsgi:app`
Manual run	`run.py`	development only

Procedure

```
cp .env.example .env
# fill in: SECRET_KEY (mandatory), ADMIN_SECRET_KEY, CHEAT_TESTER_CODE
make doctor
make compose-up
make smoke-test BASE_URL=http://localhost:5000
```

Compose fails when a mandatory secret is missing (`${VAR:?…}`) — protection against a deployment with empty keys.

Before production

1. **Rotate the secrets** if the repository was ever public: see ../security/secrets.md.
2. `SESSION_COOKIE_SECURE=true` and HTTPS at the proxy.
3. `CORS_ORIGINS` — an explicit domain list instead of `*`.
4. `DEBUG=false` and `ENV=production` (otherwise tracebacks and verbose errors).
5. `SOCKETIO_ASYNC_MODE=eventlet` for load; `threading` is the fallback without extra dependencies.
6. A single process: rooms and the scheduler live in memory. Several replicas without shared state are not supported.
7. Back up the `brainstorm-data` volume, see ../database/storage.md.

After deployment

```
make smoke-test BASE_URL=https://<domain>
curl -s https://<domain>/health | python -m json.tool
```